

ASSIGNMENT 5

EVALUATION OF CRYPTOGRAPHIC HASH FUNCTIONS AND DIGITAL SIGNATURES IN ELECTRONIC HEALTH RECORD SECURITY

Course Outcomes: CO2 and CO3

Bloom's Taxonomy Level: BL5 – Evaluate

SDG Mapping: SDG 3, SDG 9, SDG 16

STUDENT DETAILS

Austin Stejoh / 192311455

Monish Kumar / 192372073

Project Website: <https://austincrypto.freebuff.app/>

1. ABSTRACT

Electronic Health Records (EHRs) contain highly sensitive medical information including patient diagnoses, prescriptions, laboratory reports, and medical histories. Since these records are frequently exchanged between hospitals, doctors, diagnostic laboratories, and insurance providers, maintaining their integrity and authenticity is essential.

This project presents SecureEHR, a web-based security system that demonstrates the use of cryptographic hash functions and digital signatures for protecting electronic medical documents. SHA-256 and SHA3-256 are used to generate cryptographic hashes, while RSA-2048 with RSA-PSS and SHA-256 is used to generate and verify digital signatures.

The system allows users to create sample medical records, calculate their cryptographic hashes, digitally sign documents, verify signatures, and simulate unauthorized modification. The tampering experiment demonstrates that changing even a single field in an EHR produces a different hash and causes digital signature verification to fail.

The proposed solution provides integrity assurance, source authentication, non-repudiation, tamper detection, and auditability. The project demonstrates how cryptographic mechanisms can be integrated into healthcare information systems to support secure electronic health information exchange.

2. INTRODUCTION

Healthcare organizations increasingly depend on digital systems to store and exchange patient information. Electronic Health Records make medical information easily accessible to authorized healthcare professionals, but they also introduce security risks.

An attacker who modifies a diagnosis, prescription dosage, laboratory result, or patient information during transmission could potentially cause serious consequences. Therefore, healthcare systems require mechanisms to detect unauthorized modifications and verify the identity of the organization or individual that created the document.

Cryptographic hashing provides an efficient mechanism for detecting changes to digital information. A hash function converts a message into a fixed-length digest. If the original message is modified, its resulting hash will be different.

Digital signatures provide additional security by allowing a sender to sign a document using a private key. The receiver can use the corresponding public key to verify the signature.

3. PROBLEM STATEMENT

A telemedicine healthcare provider manages sensitive Electronic Health Records transmitted between medical facilities, doctors, diagnostic laboratories, and insurance portals. The healthcare network must ensure that patient diagnoses and prescription documents remain unaltered during transit and that the originating healthcare provider's identity is cryptographically verified.

Therefore, a cryptographic system using secure hash algorithms and digital signatures is required to provide reliable integrity, authentication, non-repudiation, and auditability for EHR exchange.

4. OBJECTIVES

1. Implement SHA-256 hashing for medical documents.
2. Implement SHA3-256 hashing for comparison.
3. Generate RSA-2048 public and private keys.
4. Digitally sign medical documents using RSA-PSS.
5. Verify digital signatures.
6. Demonstrate the effect of unauthorized data modification.
7. Compare cryptographic algorithms based on security and performance.
8. Maintain an audit trail of security operations.
9. Design an end-to-end EHR integrity and authentication system.
10. Recommend a suitable cryptographic deployment for healthcare information exchange.

5. SCOPE OF THE PROJECT

- Medical record creation
- SHA-256 and SHA3-256 hash generation
- RSA key generation
- Digital signature creation
- Signature verification
- Tamper testing
- Algorithm comparison
- Security architecture visualization
- Audit logging

The application uses synthetic patient data only and is intended for educational and demonstration purposes.

6. PROPOSED SYSTEM

The proposed system, named SecureEHR, is a web-based cryptographic security platform. The frontend provides interfaces for EHR document creation, hash analysis, digital signing, verification, tamper testing, algorithm comparison, security architecture, and audit logs. The backend performs cryptographic operations using standard, trusted libraries.

7. DEPLOYED WEBSITE

The SecureEHR application was implemented as a functional web-based demonstration for cryptographic integrity and digital signature verification. The deployed project can be accessed at:

<https://austincrypto.freebuff.app/>

The website demonstrates the project workflow through a user interface for medical-document hashing, digital signatures, verification, and tamper detection. It is intended for educational demonstration and uses synthetic patient information.

8. SYSTEM ARCHITECTURE

Healthcare Provider

|

SecureEHR – Cryptographic Integrity & Authentication Platform

```

      v
Medical Record
      |
      v
SHA-256 / SHA3-256
      |
      v
Hash Digest
      |
      v
RSA Private Key
      |
      v
Digital Signature
      |
      v
Secure Transmission
      |
      v
Receiving Healthcare Facility
      |
      v
RSA Public Key / Certificate
      |
      v
Signature Verification
      |
      v
Integrity + Authentication
      |
      v
Audit Log

```

9. TECHNOLOGIES USED

Technology	Purpose
React	Frontend development
TypeScript	Type-safe frontend programming
Tailwind CSS	User interface styling
FastAPI	Backend API development
Python	Cryptographic implementation
Cryptography Library	RSA and digital signature operations
SHA-256	Cryptographic hashing
SHA3-256	Cryptographic hashing
RSA-2048	Public-key cryptography
RSA-PSS	Digital signature scheme
REST API	Frontend-backend communication

10. CRYPTOGRAPHIC HASH FUNCTIONS

A cryptographic hash function transforms input data into a fixed-size output called a hash digest.

$$H(M) = \text{Hash}(M)$$

If the medical document is modified, even by one character, the resulting digest changes significantly due to the avalanche effect. This makes cryptographic hashes useful for detecting unauthorized changes.

11. SHA-256

SHA-256 is a member of the SHA-2 family and generates a 256-bit digest. Its hexadecimal representation contains 64 characters. It provides strong resistance against practical preimage, second-preimage, and collision attacks and is widely supported.

12. SHA3-256

SHA3-256 belongs to the SHA-3 family and uses a sponge construction. It also produces a 256-bit digest and provides a modern alternative hash design to SHA-2.

13. SHA-256 AND SHA3-256 COMPARISON

Parameter	SHA-256	SHA3-256
Family	SHA-2	SHA-3
Digest Size	256 bits	256 bits
Construction	Merkle-Damgård	Sponge
Collision Resistance	Strong	Strong
Preimage Resistance	Strong	Strong
Adoption	Very High	High
Performance	Excellent	Good
EHR Suitability	Excellent	Excellent

14. DIGITAL SIGNATURES

A digital signature is a cryptographic mechanism used to verify the authenticity and integrity of digital information. The sender hashes the document and creates a signature using a private key. The receiver uses the corresponding public key to verify it.

$$H = \text{SHA256}(M)$$
$$S = \text{Sign}(H, \text{PrivateKey})$$

Verification succeeds only when the signature corresponds to the received document and the public key.

15. RSA-2048 WITH RSA-PSS

The proposed system uses RSA-2048 with RSA-PSS and SHA-256. RSA uses a private key for signing and a public key for verification. The private key must remain confidential. RSA-PSS provides modern randomized padding for RSA signatures.

16. SAMPLE EHR DATA

Patient ID: P10245
Patient Name: Arun Kumar
Diagnosis: Type 2 Diabetes
Prescription: Metformin 500 mg

```
Doctor ID: DOC4582
Hospital: ABC Healthcare
Date: 01-09-2026
```

17. HASH GENERATION PROCESS

```
Medical Record
  |
  +----> SHA-256
  |
  +----> SHA3-256
  |
  v
Hash Digests
```

The system displays the hash algorithm, digest, hash length, and processing time.

18. DIGITAL SIGNATURE CREATION

```
Medical Document
  |
  v
SHA-256
  |
  v
RSA-PSS + Private Key
  |
  v
Digital Signature
```

19. DIGITAL SIGNATURE VERIFICATION

The receiver obtains the medical document, digital signature, and the verified public key or certificate. The receiver calculates the hash of the received document and verifies the signature.

```
VALID:
  Signature valid
  Document integrity confirmed
  Provider identity authenticated

INVALID:
  Signature invalid
  Possible document modification
  Possible invalid or compromised key
```

20. TAMPER DETECTION EXPERIMENT

The following experiment changes only the prescription dosage.

```
Original:
Prescription: Metformin 500 mg

Modified:
Prescription: Metformin 1000 mg
```

The modified record generates a different cryptographic hash. The original digital signature is then tested against the modified document.

```
HASH MATCH: NO
SIGNATURE STATUS: INVALID
UNAUTHORIZED MODIFICATION DETECTED
```

21. EXPERIMENTAL TEST CASES

Test Case	Modification	Expected Result
1	Original document	Signature Valid
2	Change patient ID	Signature Invalid
3	Change diagnosis	Signature Invalid
4	Change prescription	Signature Invalid
5	Modify signature	Verification Failed
6	Modify document and signature without valid private key	Verification Failed
7	Original document + original signature	Signature Valid

22. SECURITY ANALYSIS

21.1 Integrity

Hash functions detect whether a document has changed. If the expected digest and the digest calculated from the received content differ, the document has been modified or corrupted.

21.2 Authentication

Digital signatures help verify that a document was signed using the private key corresponding to a verified public key. Certificate-based PKI is required to associate the key with a trusted healthcare provider identity.

21.3 Non-Repudiation

Digital signatures provide evidence that a particular private key was used to sign a document. Effective non-repudiation also depends on secure private-key management, identity verification, certificate management, and reliable audit records.

21.4 Collision Resistance

A collision occurs when two different messages produce the same hash. SHA-256 and SHA3-256 are designed to make practical collision attacks computationally infeasible.

21.5 Tamper Detection

Any unauthorized modification of a signed EHR causes signature verification to fail, providing a strong mechanism for detecting suspicious documents.

23. PERFORMANCE AND COMPUTATIONAL OVERHEAD

Operation	Relative Overhead
SHA-256 Hash	Very Low
SHA3-256 Hash	Low
RSA Signature Generation	Moderate
RSA Signature Verification	Moderate
Audit Logging	Low

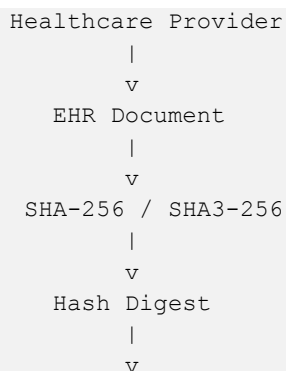
24. SECURITY THREATS AND MITIGATION

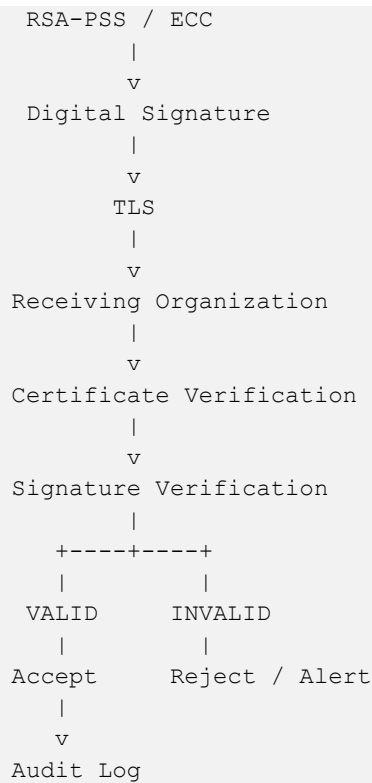
Threat	Mitigation
Document modification	Hash + digital signature
Provider impersonation	Digital signature + PKI
Signature modification	Signature verification
Hash collision attack	SHA-256 / SHA3-256
Private key theft	Secure key storage
Network interception	TLS
Unauthorized access	Authentication + access control
Audit manipulation	Secure/tamper-resistant logging
Replay of old messages	Timestamps, nonces, and document versioning

25. LIMITATIONS

11. Hashing and digital signatures do not provide confidentiality.
12. Private-key compromise can undermine signature security.
13. Public-key authentication requires trusted certificate management.
14. The prototype uses synthetic medical records rather than production EHR data.
15. A real healthcare deployment requires additional access-control and compliance mechanisms.
16. Secure audit logging must be protected against unauthorized modification.
17. Key rotation and revocation must be implemented for production environments.

26. RECOMMENDED SECURITY ARCHITECTURE





27. RECOMMENDED PROTOCOL DEPLOYMENT

- Document Integrity: SHA-256 or SHA3-256
- Digital Signature: RSA-PSS with SHA-256 or an appropriately standardized elliptic-curve signature scheme
- Secure Communication: TLS
- Authentication: PKI and digital certificates
- Auditability: Secure audit logs containing document ID, provider identity, timestamp, operation, verification result, and key/certificate identifier

28. RESULTS

18. SHA-256 successfully generates a 256-bit digest.
19. SHA3-256 successfully generates a 256-bit digest.
20. RSA-2048 key pairs can be generated for signing and verification.
21. Medical documents can be digitally signed.
22. Valid signatures can be successfully verified.
23. Modification of medical information changes the document hash.
24. Modified documents fail digital signature verification.
25. The system can identify potential tampering.
26. Cryptographic operations can be integrated into an EHR workflow.
27. Audit logs provide additional accountability.

29. ADVANTAGES

- Strong data integrity
- Provider authentication
- Digital document verification
- Tamper detection
- Non-repudiation support
- Efficient hashing
- Scalable architecture
- Auditability
- Secure healthcare information exchange

30. FUTURE ENHANCEMENTS

28. Blockchain-based audit trails.
29. Hardware Security Module (HSM) integration.
30. Cloud-based key management.
31. Multi-factor authentication.
32. Role-Based Access Control (RBAC).
33. Attribute-Based Access Control (ABAC).
34. Automated certificate revocation checking.
35. Secure EHR APIs based on healthcare interoperability standards.
36. Real-time security alerts.
37. Advanced anomaly and intrusion detection.
38. Encryption for confidentiality.
39. Post-quantum cryptographic algorithms for future resilience.

31. CONCLUSION

The SecureEHR project demonstrates the practical application of cryptographic hash functions and digital signatures in Electronic Health Record security.

SHA-256 and SHA3-256 provide strong mechanisms for detecting modifications to medical documents. RSA-2048 with RSA-PSS provides digital signatures that enable document integrity verification and source authentication when combined with trusted public-key infrastructure.

The tampering experiment clearly demonstrates that modifying even a small portion of a medical record changes its cryptographic representation and causes signature verification to fail.

The evaluation shows that the combination of secure hashing, digital signatures, TLS, PKI, access control, secure key management, and audit logging can provide a strong foundation for secure health information exchange.

Therefore, cryptographic mechanisms are essential for maintaining integrity, authentication, accountability, and trust in modern electronic healthcare systems.

32. REFERENCES

40. National Institute of Standards and Technology (NIST), Secure Hash Standard (SHS), FIPS 180-4.
41. National Institute of Standards and Technology (NIST), SHA-3 Standard: Permutation-Based Hash and Extendable-Output Functions, FIPS 202.

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43. National Institute of Standards and Technology (NIST), Recommendation for Key Management.
44. William Stallings, Cryptography and Network Security: Principles and Practice.
45. Behrouz A. Forouzan, Cryptography and Network Security.
46. Python Cryptography Library Documentation.
47. FastAPI Documentation.
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APPENDIX A – SAMPLE PYTHON IMPLEMENTATION

```
import hashlib
from cryptography.hazmat.primitives.asymmetric import rsa, padding
from cryptography.hazmat.primitives import hashes
from cryptography.exceptions import InvalidSignature

medical_record = """
Patient ID: P10245
Patient Name: Arun Kumar
Diagnosis: Type 2 Diabetes
Prescription: Metformin 500 mg
Doctor ID: DOC4582
Hospital: ABC Healthcare
Date: 01-09-2026
"""

sha256_hash = hashlib.sha256(medical_record.encode()).hexdigest()
sha3_hash = hashlib.sha3_256(medical_record.encode()).hexdigest()

private_key = rsa.generate_private_key(
    public_exponent=65537,
    key_size=2048
)
public_key = private_key.public_key()

signature = private_key.sign(
    medical_record.encode(),
    padding.PSS(
        mgf=padding.MGF1(hashes.SHA256()),
        salt_length=padding.PSS.MAX_LENGTH
    ),
    hashes.SHA256()
)

try:
    public_key.verify(
        signature,
        medical_record.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("Signature Verification: SUCCESS")
except InvalidSignature:
```

```
print("Signature Verification: FAILED")
```

APPENDIX B – TAMPERING TEST

```
modified_record = medical_record.replace(
    "Metformin 500 mg",
    "Metformin 1000 mg"
)

try:
    public_key.verify(
        signature,
        modified_record.encode(),
        padding.PSS(
            mgf=padding.MGF1(hashes.SHA256()),
            salt_length=padding.PSS.MAX_LENGTH
        ),
        hashes.SHA256()
    )
    print("Signature Verification: SUCCESS")
except InvalidSignature:
    print("Signature Verification: FAILED")
```

WEBSITE ACCESS

Project Website: <https://austincrypto.freebuff.app/>

The deployed SecureEHR website provides the practical interface for demonstrating cryptographic hashing, digital signature generation and verification, and unauthorized document modification detection.

PROJECT TEAM

Student Name	Register Number
Austin Stejoh	192311455
Monish Kumar	192372073